

ITI AI-Pro

Natural Language Processing



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- Output Decoding

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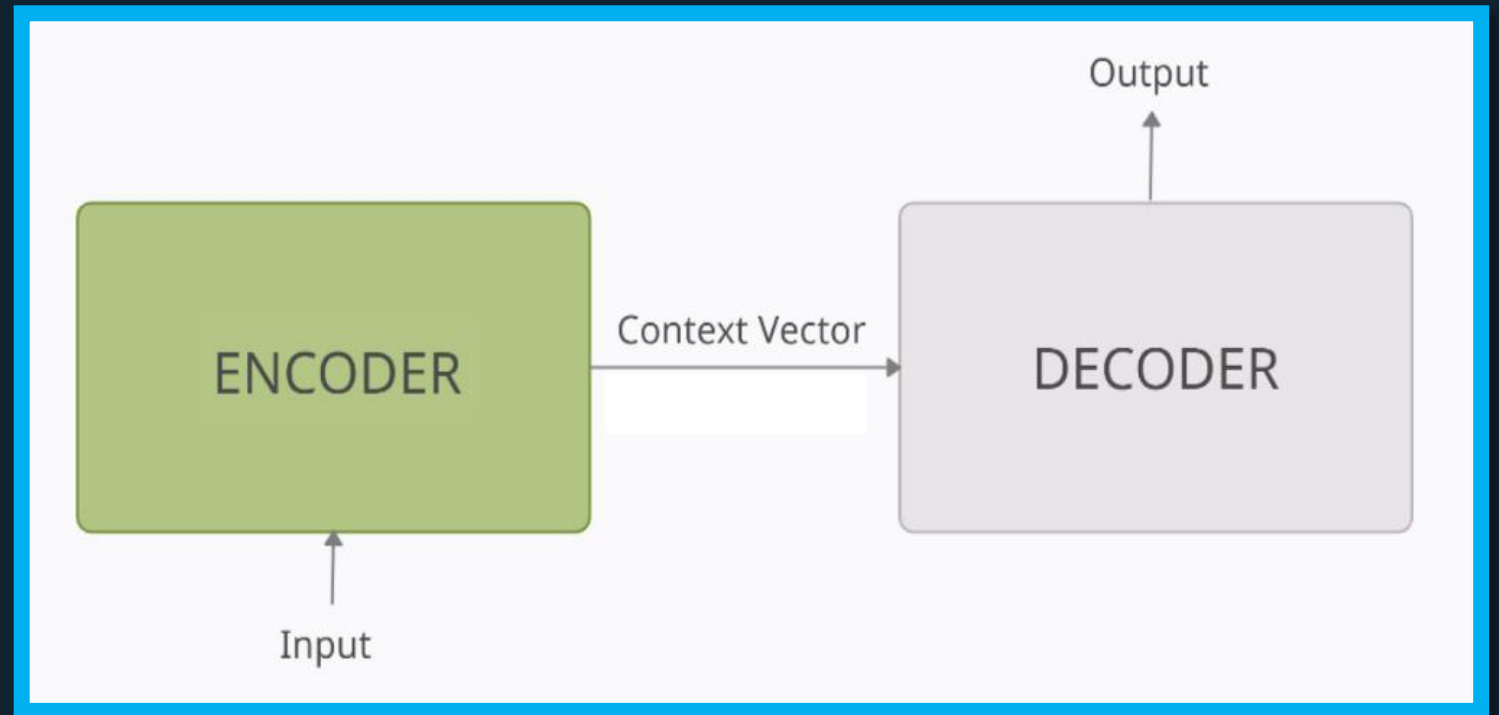
Part I

Seq 2 Seq Models



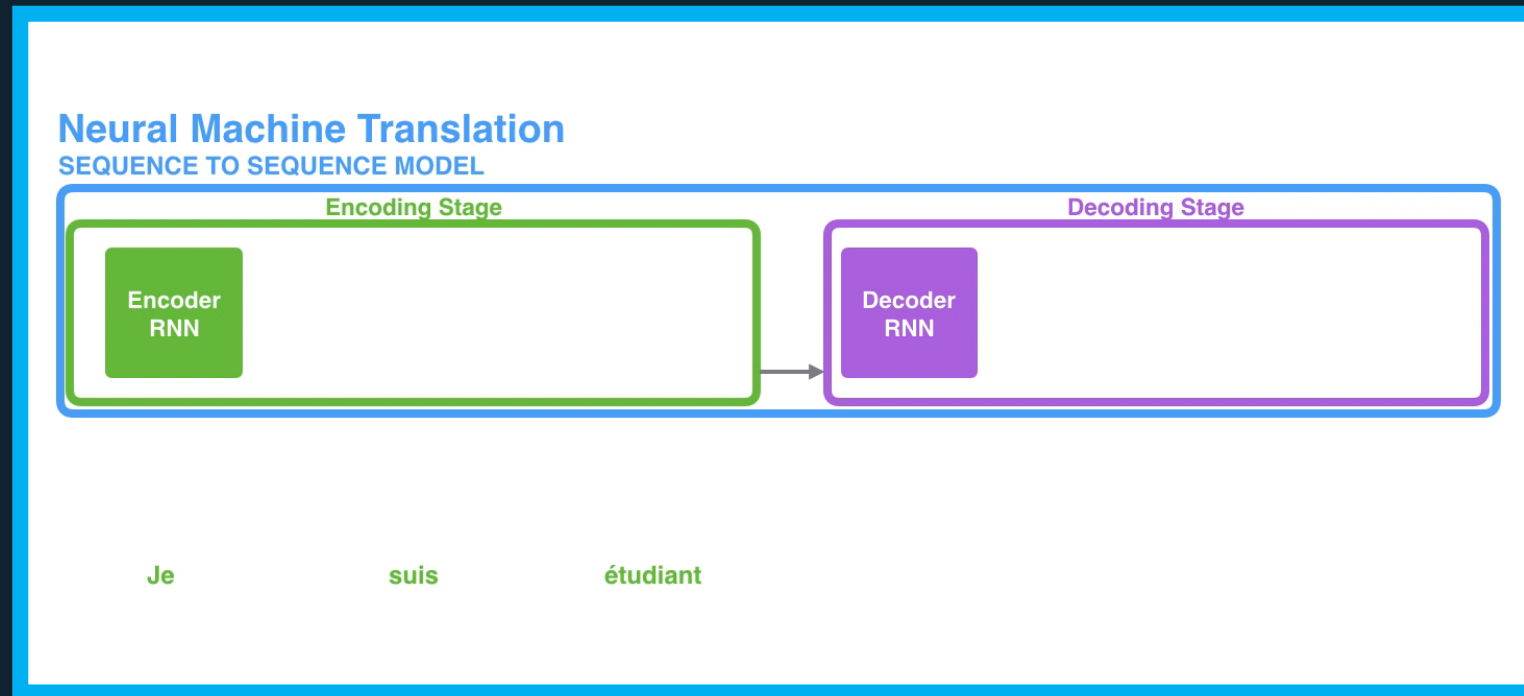
Seq 2 Seq

- Encoders and Decoders can be **different** kind of **architecture** such as: RNN, LSTM, BERT ... etc.



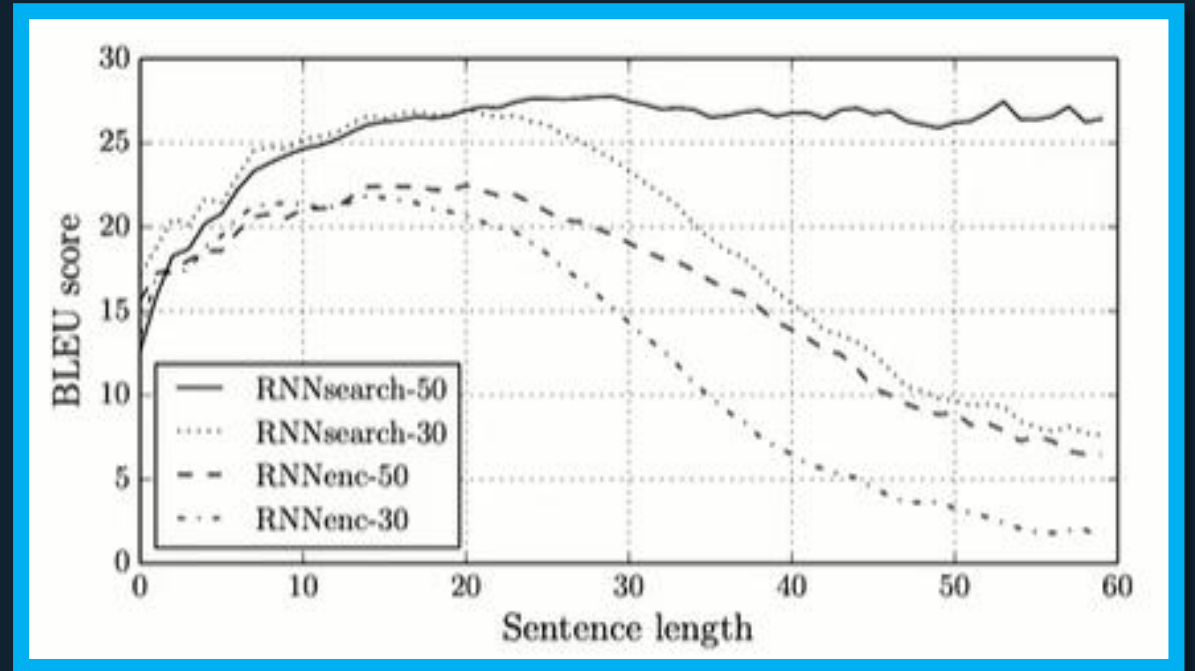
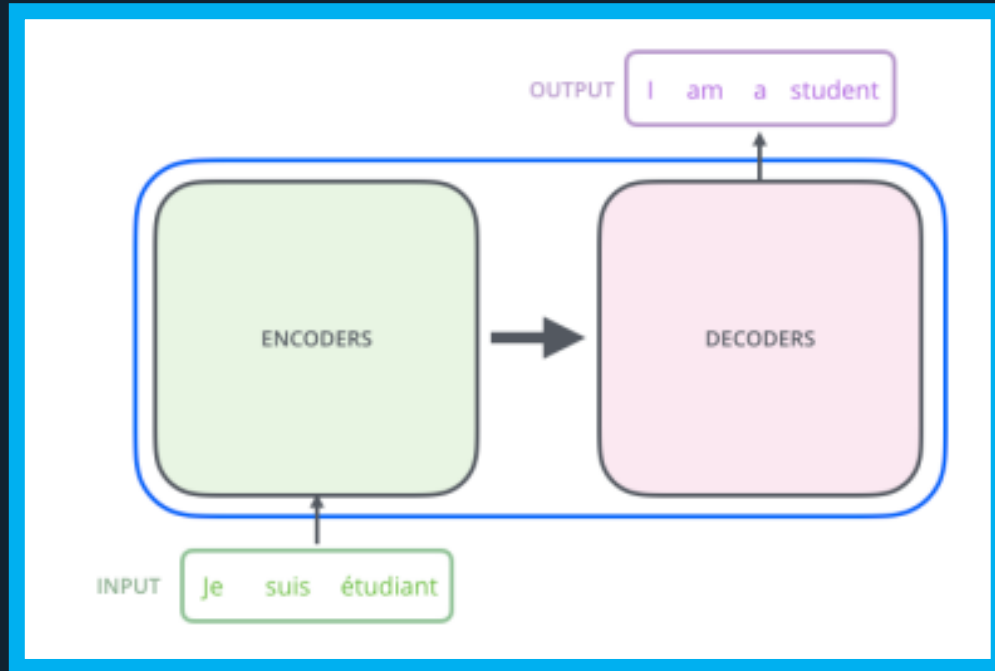
Seq 2 Seq

- **Encoders** : encode (extract) all info necessary for the process
- **Decoders** : reconstruct context vector to the desired output



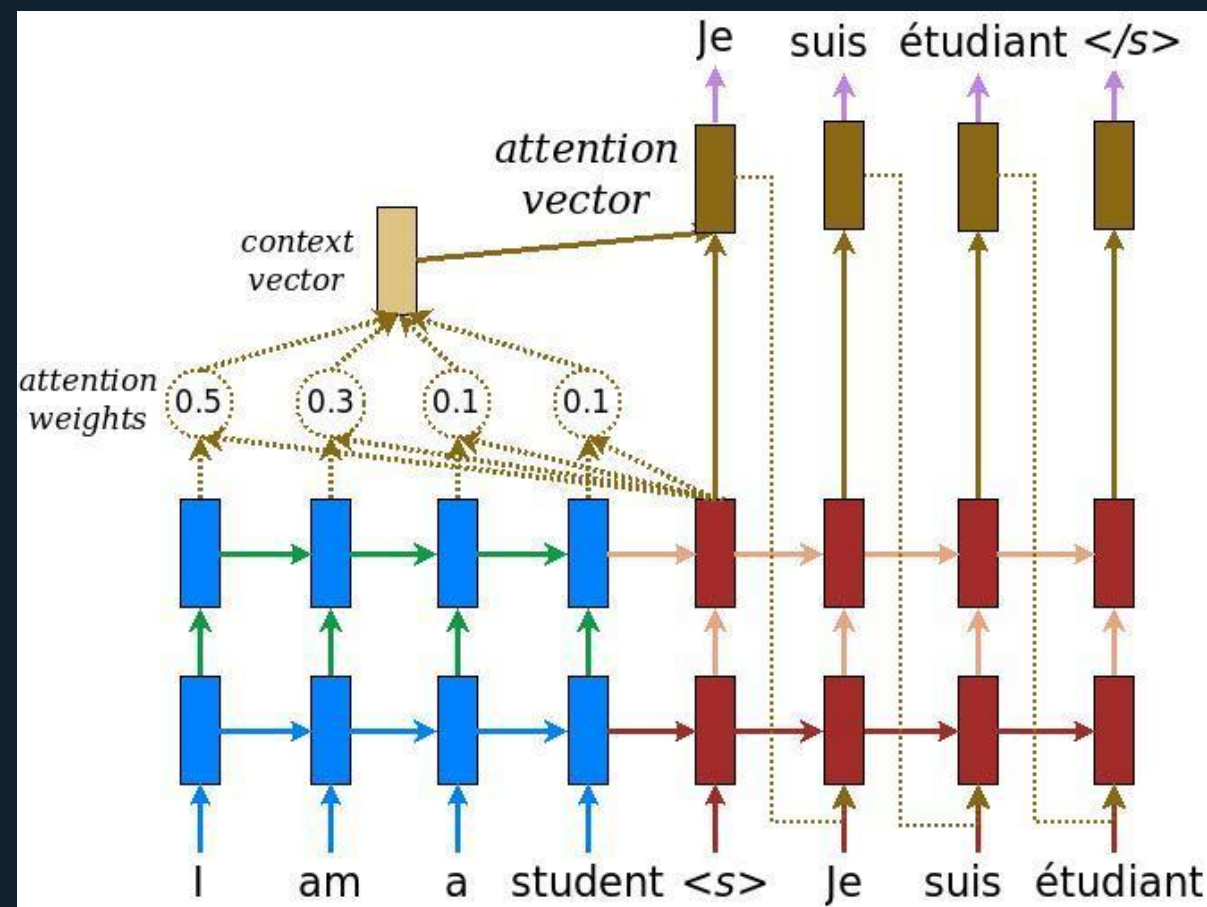
NMT

- **NMT** : Neural Machine Translation
- Large seq → Attention layer

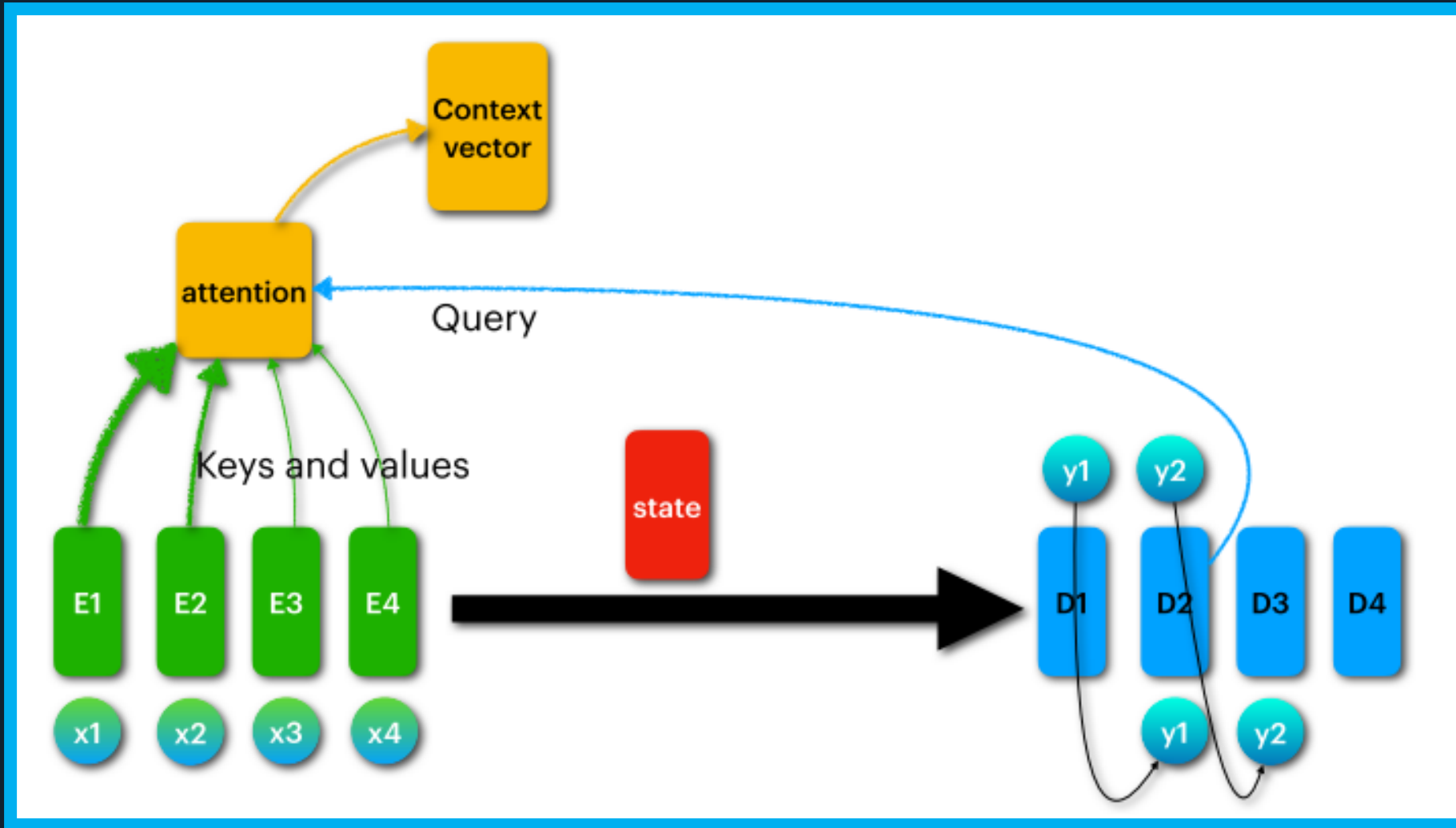


ATTENTION-BASED NMT

- Attention **supports** the **decoder output at every prediction** through focusing on **what matters** each one.
- The decoder uses attention to **build a context vector** to **generate the output logit**.
- **Every output step** of the decoder is **fed back** to the **next step** as input.



ATTENTION-BASED NMT



ATTENTION-BASED NMT

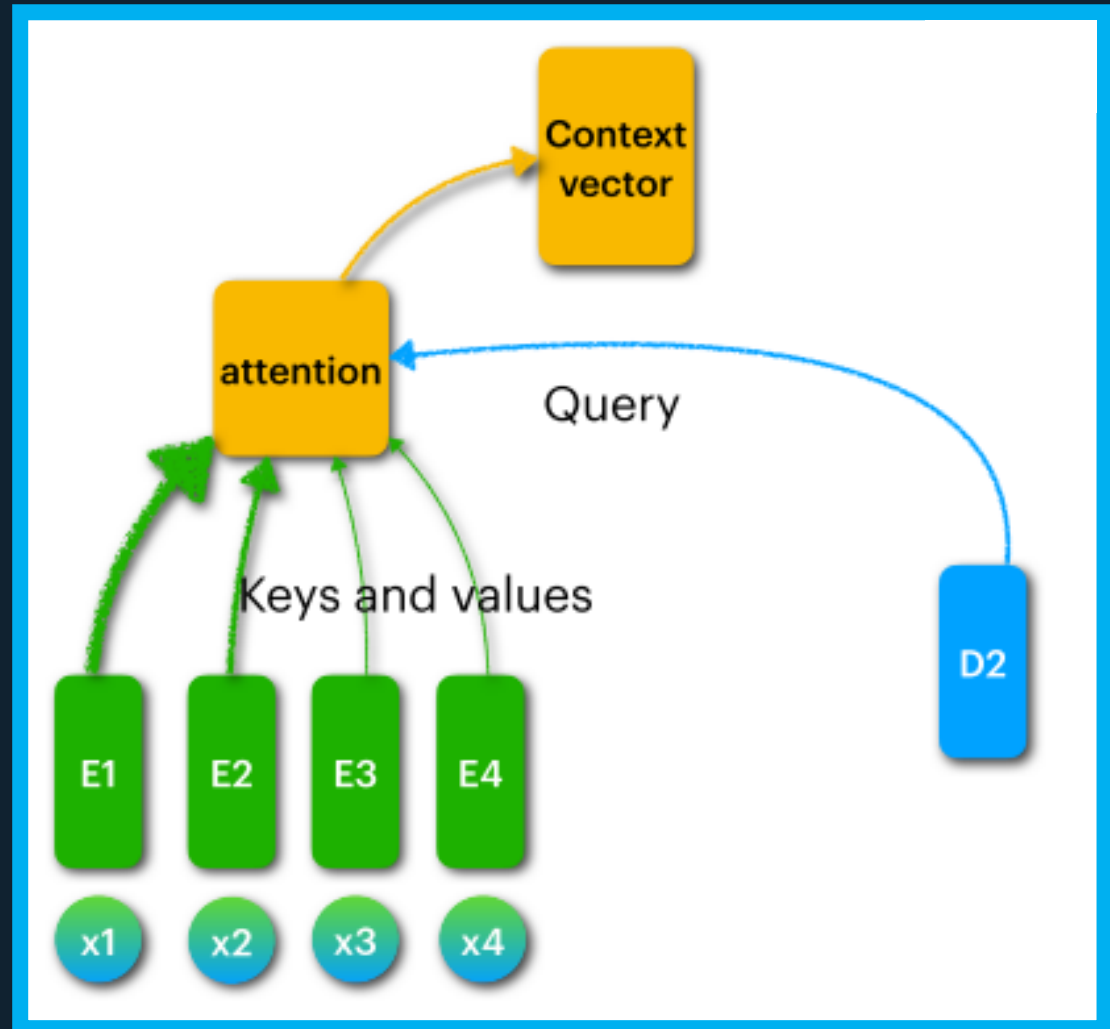
Working KH

$$\alpha_{de} = \frac{\exp(\text{score}(h_e, h_d))}{\sum_{e'=1}^e \exp(\text{score}(h_{e'}, h_d))}$$

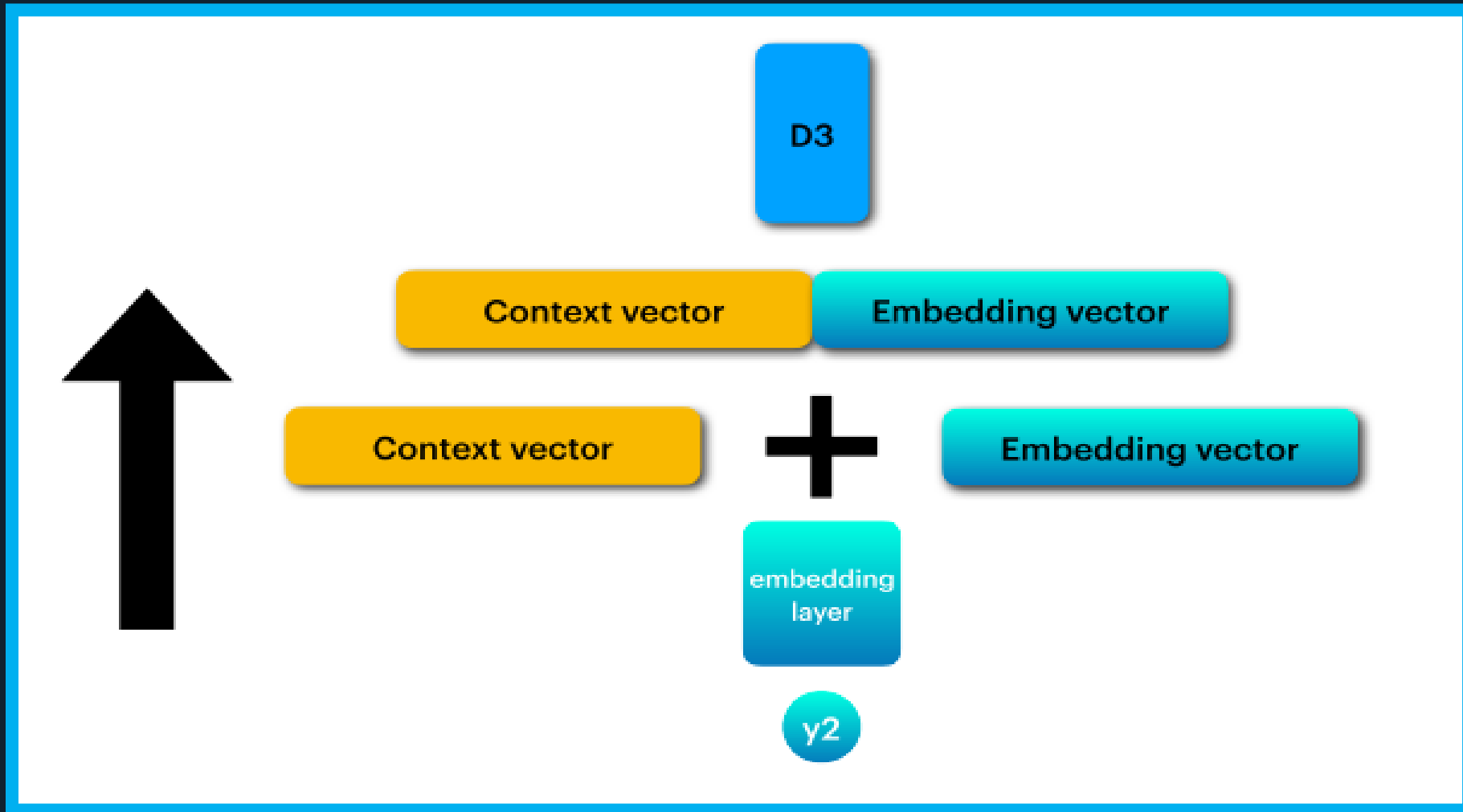
Attention weights

$$c_t = \sum_e \alpha_{de} h_e$$

Context vector



ATTENTION-BASED NMT

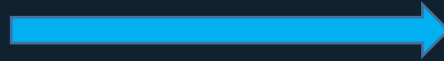




NMT EVALUATION

Predicted Translation
(candidate)

compare



True Translation
(reference)

BLEU

ROUGE-N

F1

NMT EVALUATION

- **BLEU Score:** BiLingual Evaluation Understudy

candidate	I	I	am	I	
reference 1	Younes	said	I	am	hungry
reference 2	He	said	I	am	hungry
Count:	1				

NMT EVALUATION

- BLEU Modified Score

candidate	I	I	am	I	
reference 1	Younes	said	I	am	hungry
reference 2	He	said	I	am	hungry
Count:			1		

NMT EVALUATION

- **ROUGE-N Score:** Recall-Oriented Understudy for Gisting Evaluation

candidate	I	I	am	I	
reference 1	Younes	said	I	am	hungry
reference 2	He	said	I	am	hungry
	Count 1:		2/5	Count 2:	
				2/5	

NMT EVALUATION

- F1 - Score

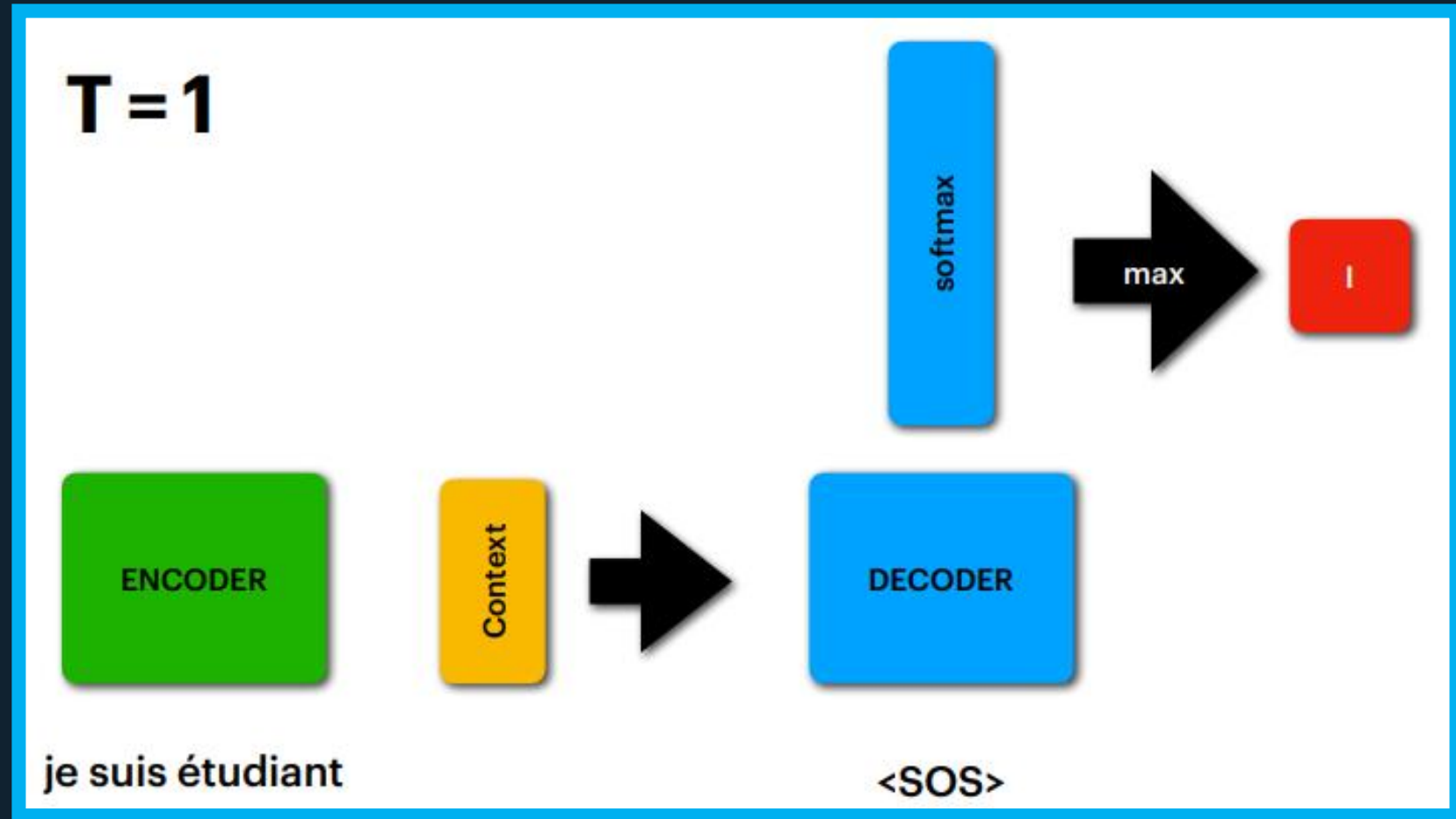
$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$F1 = 2 \times \frac{BLEU \times ROUGE-N}{BLEU + ROUGE-N}$$

$$F1 = 2 \times \frac{0.5 \times 0.4}{0.5 + 0.4} = \frac{4}{9} \approx 0.44$$

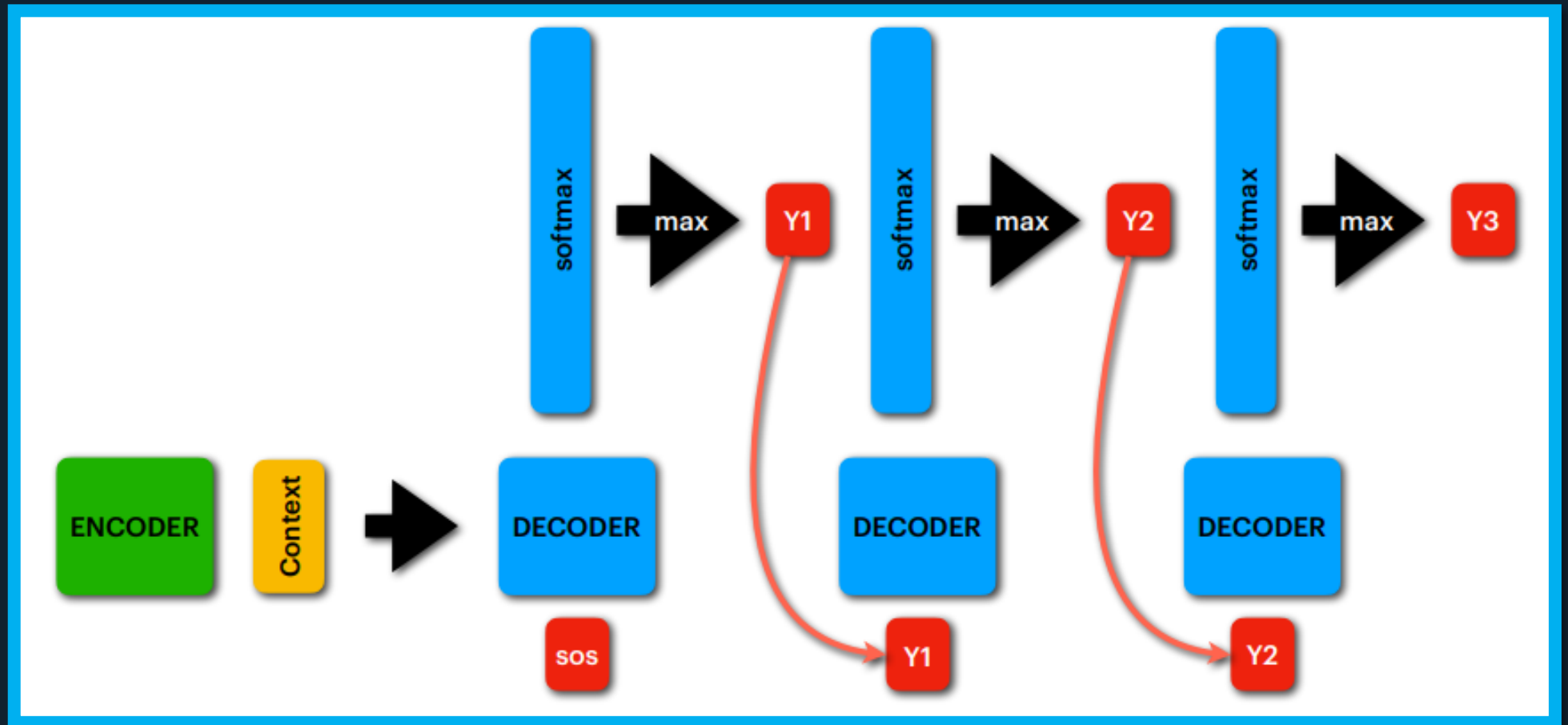
OUTPUT DECODING

- Simply,



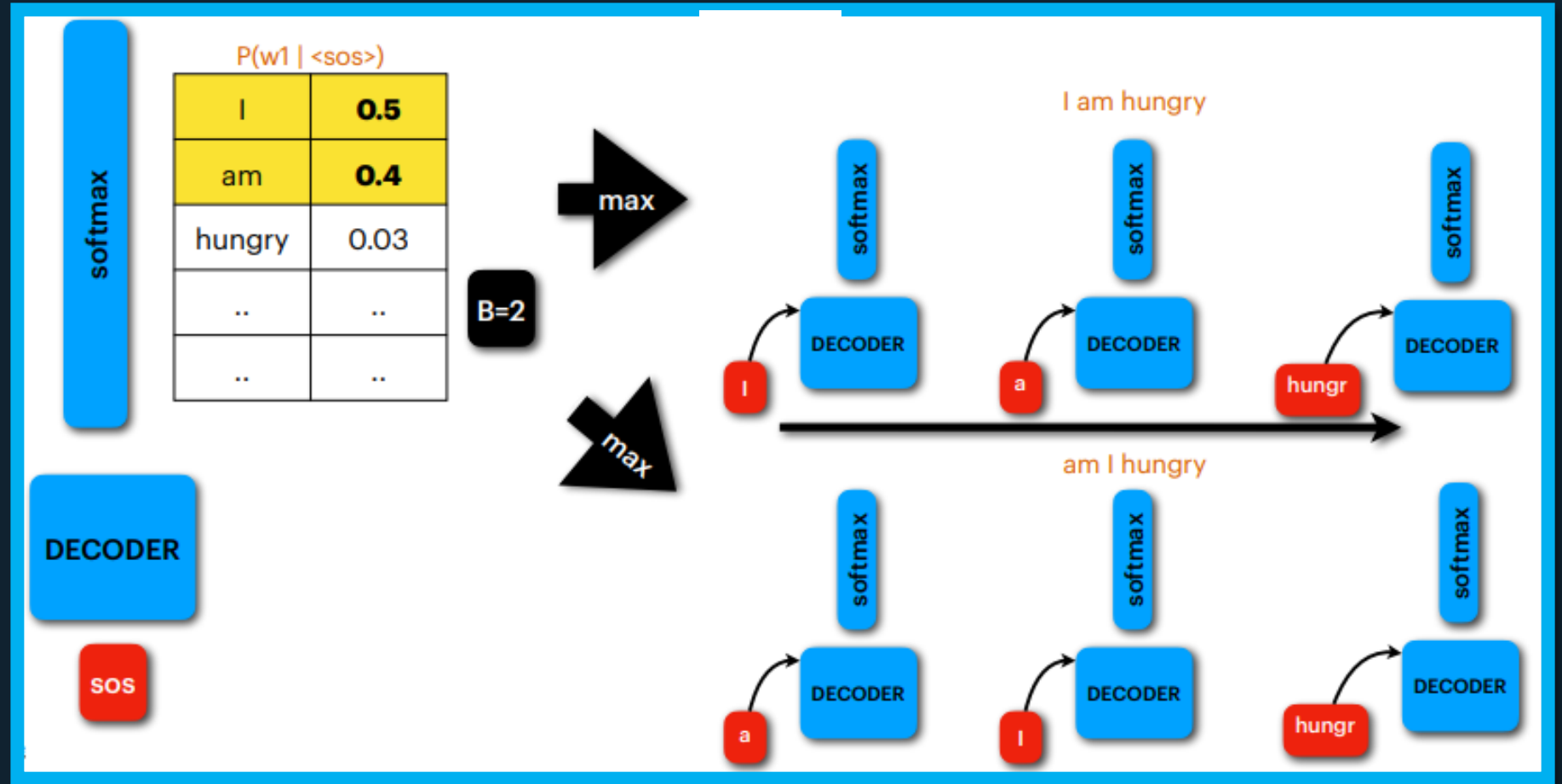
OUTPUT DECODING

- Greedy Search



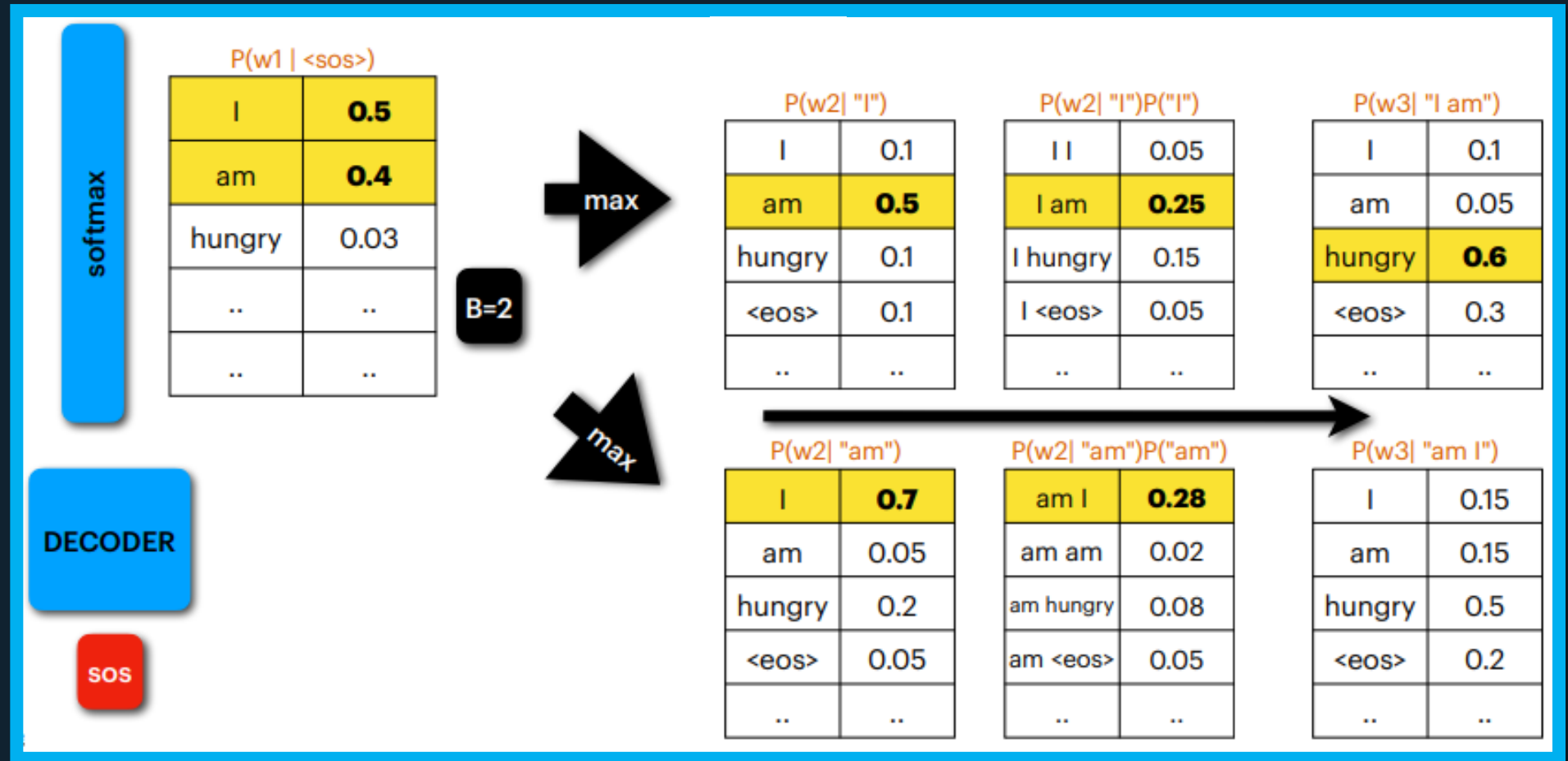
OUTPUT DECODING

- Beam Search



OUTPUT DECODING

- Beam Search



OUTPUT DECODING

	Greedy Search	Beam Search
Selection Criteria	Best Output @ <u>Each Step</u>	Best <u>Overall</u> Output
No. of Paths	1	B paths
Accuracy		
Computational Power		



NB I
NMT



Part II

Transformers



Information Bottleneck

+

Series Not Parallel

+

No Both Directions

TRANSFORMERS

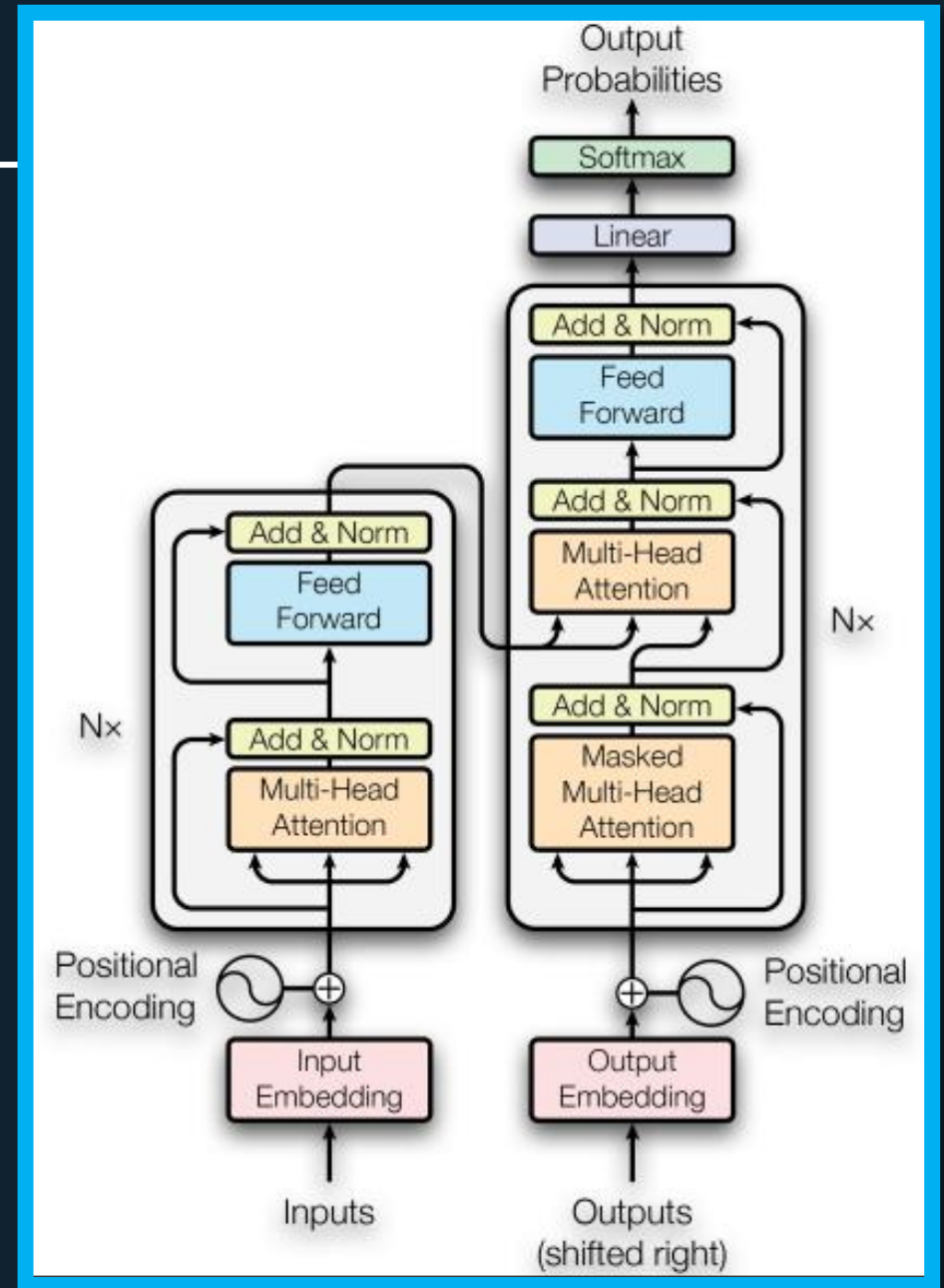
- Google 2017 : [Attention is all you need](#)
- No RNNs → Attentions

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

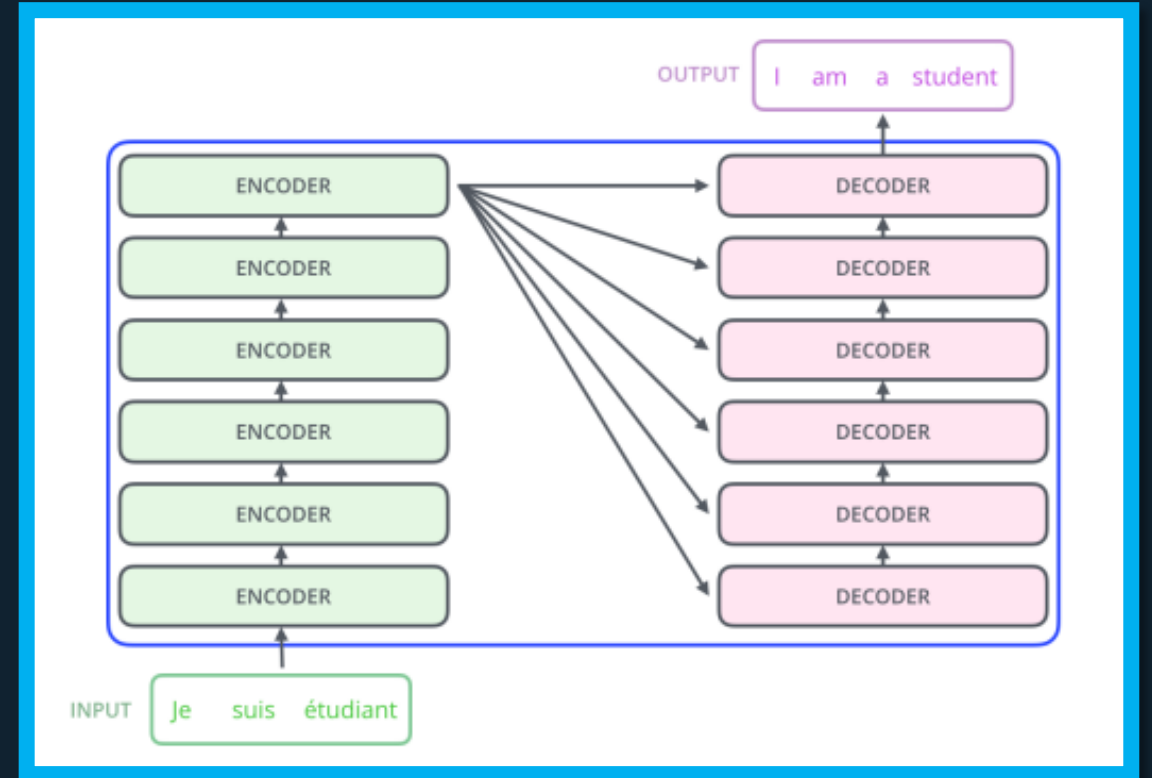
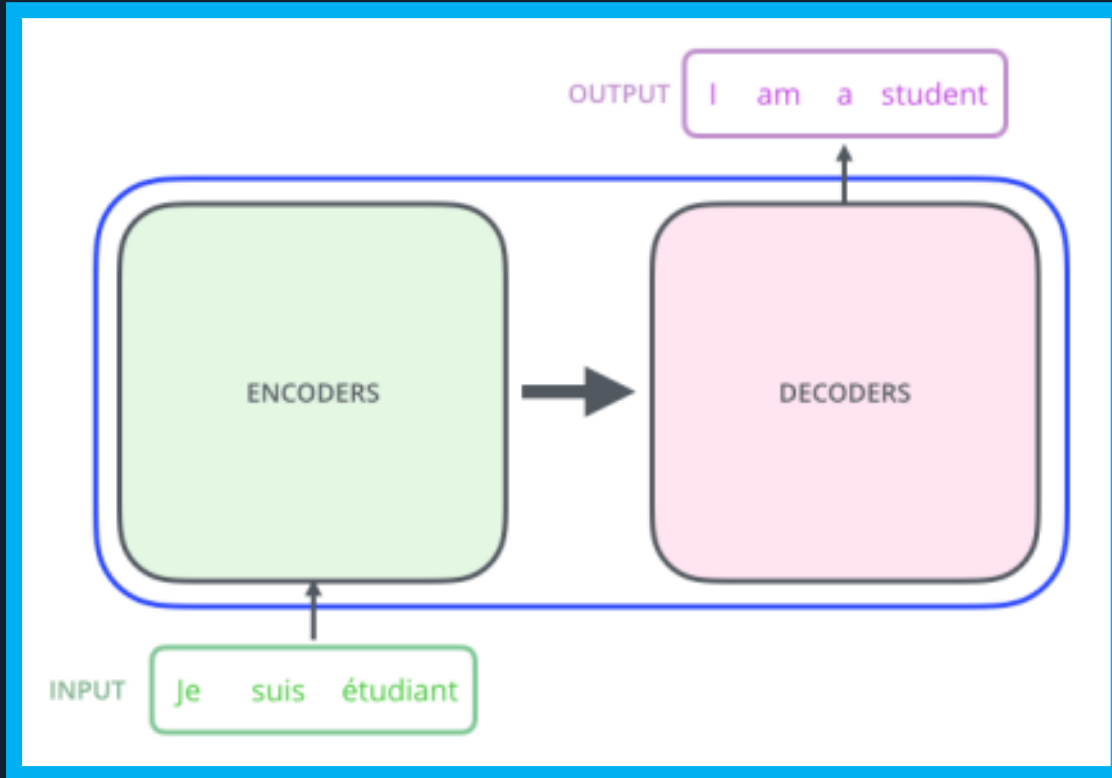
Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

ARCHITECTURE

- **Positional encoding** is used to addresses the **position**.
- **Multi-head attention** is used to learn **context** of words.
- **All the components** can be **parallelized** and thus make use of large hardware/data.



WORKING KH



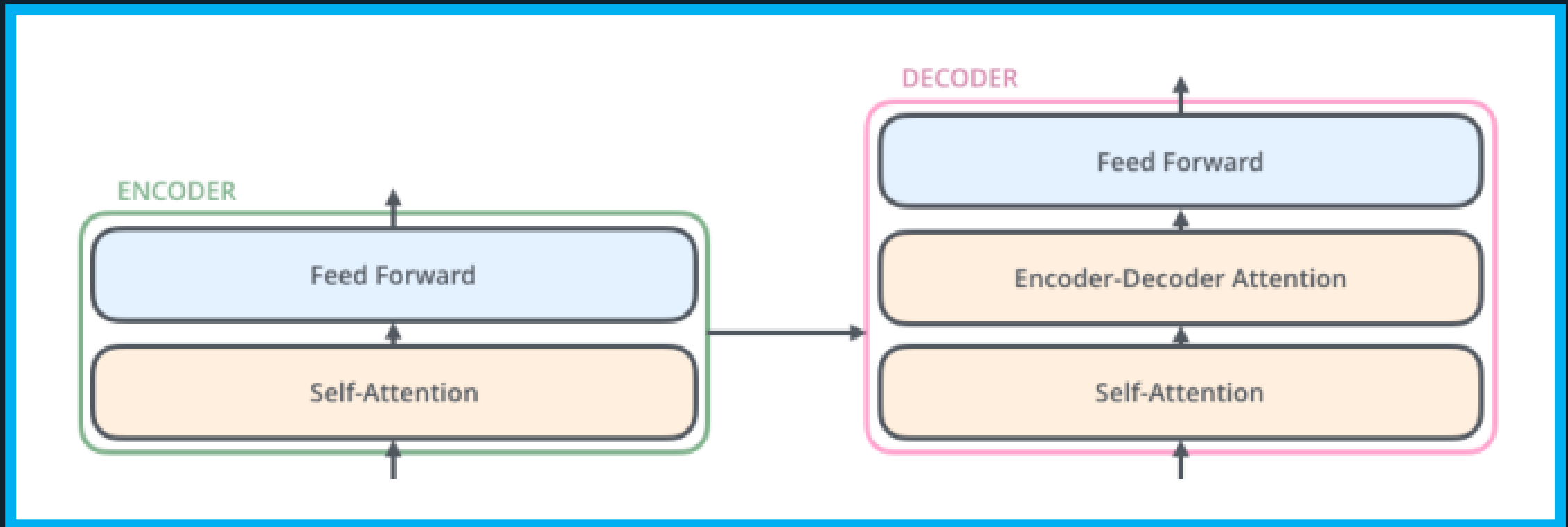
More

WORKING KH

Encoder

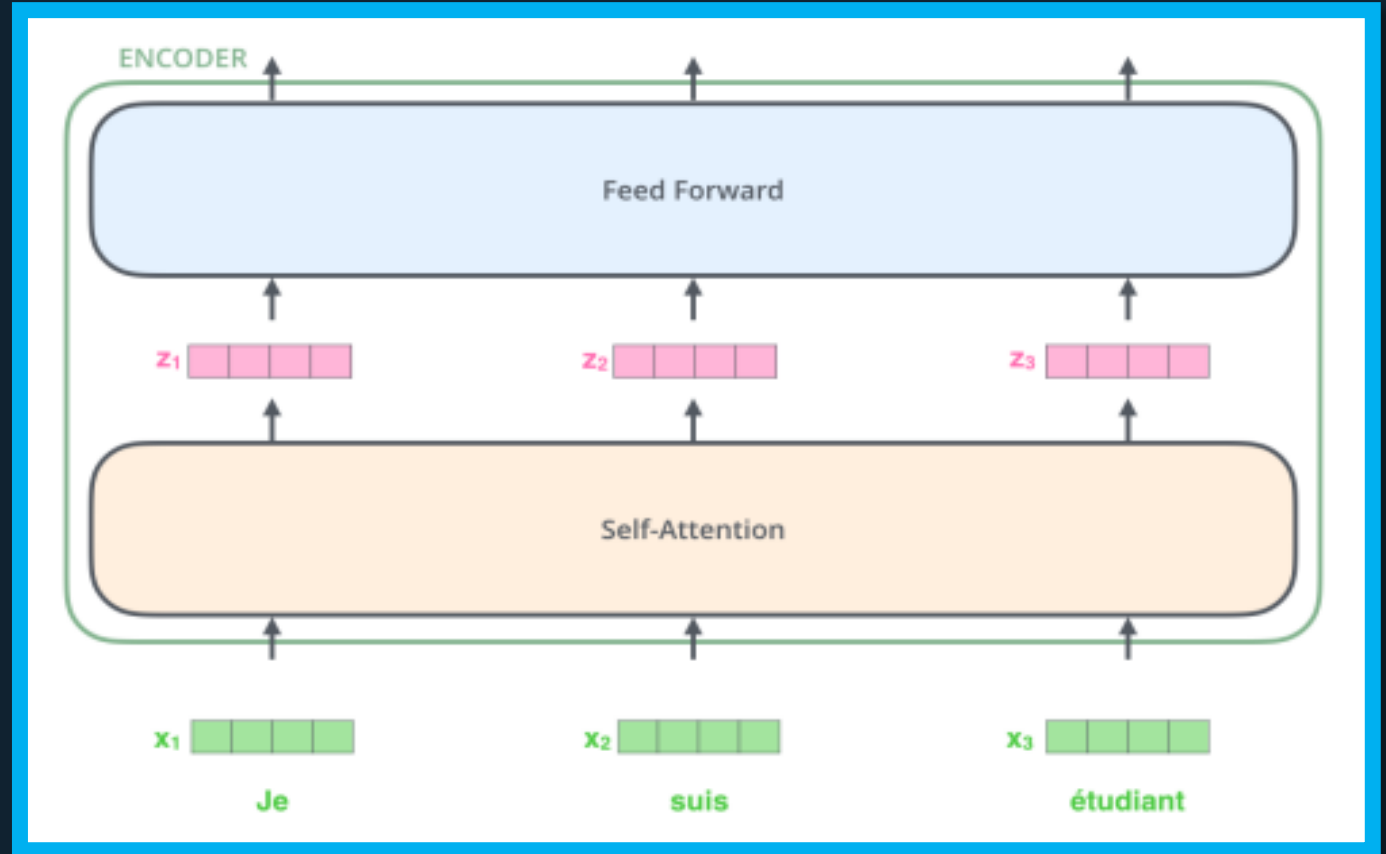
VS

Decoder



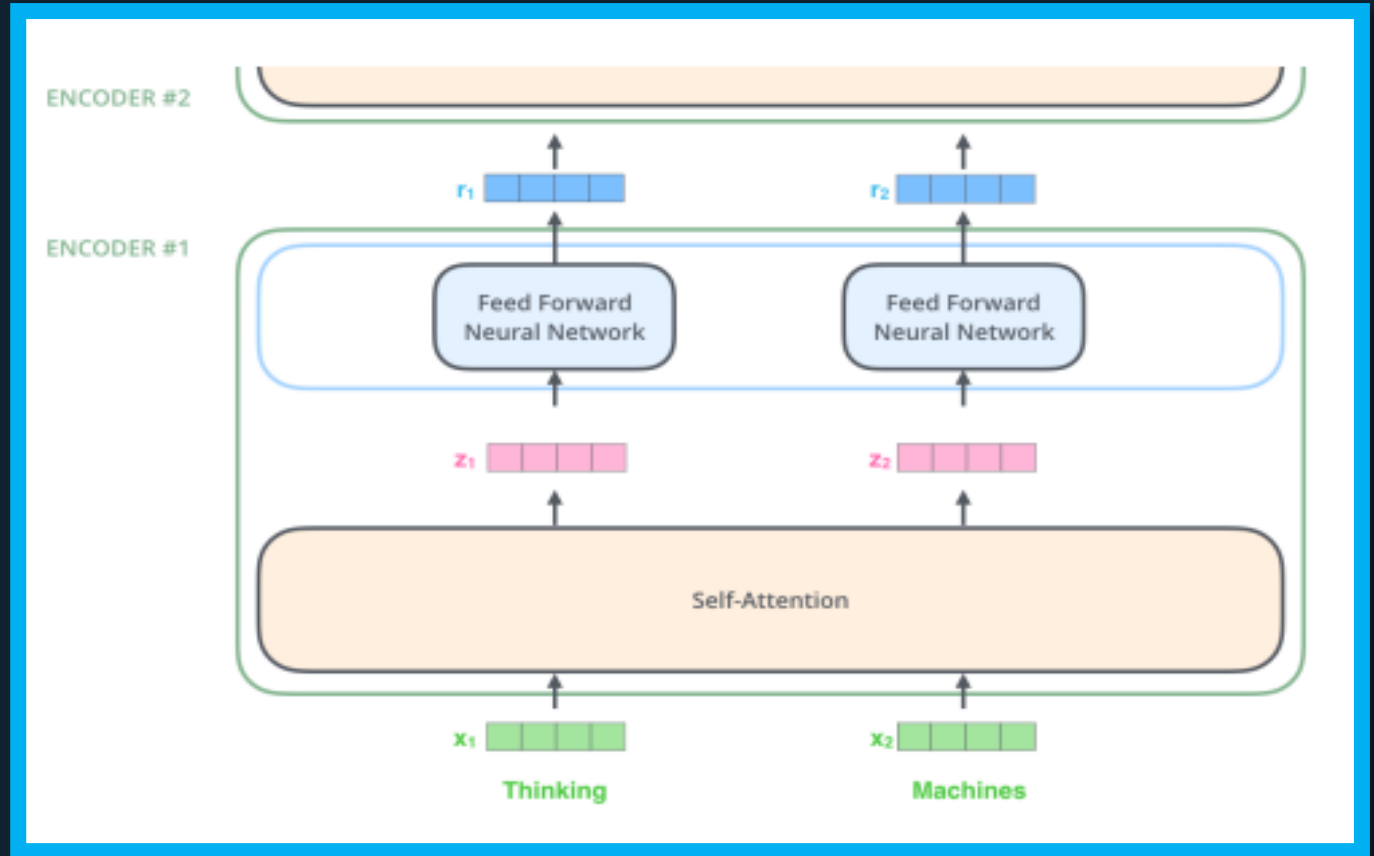
WORKING KH

- The **first encoder input** is the **word embeddings** for the input words.



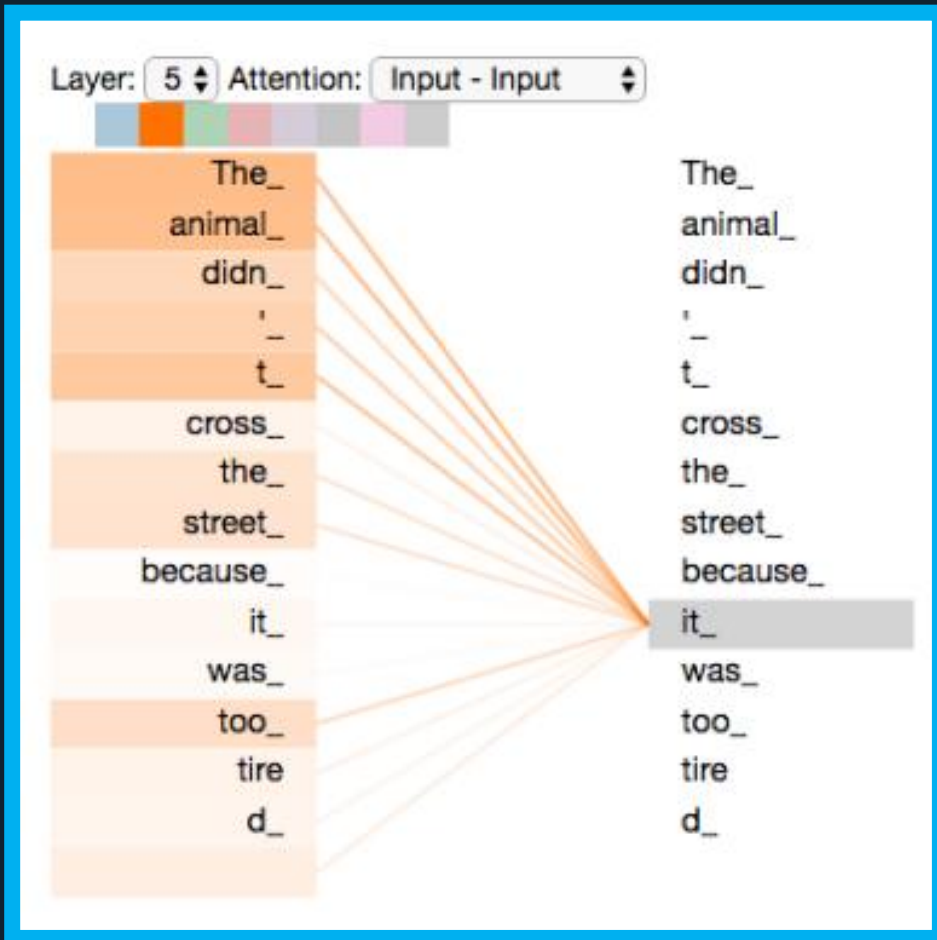
WORKING KH

- Each word goes to its **own path**, **no dependency** is required between each word and it's previous one, which makes it **easy to parallelize**.



WORKING KH

- Self-Attention



Input

Embedding

Queries

Keys

Values

Score

Divide by 8 ($\sqrt{d_k}$)

Softmax

Softmax

X
Value

Sum

Thinking

x_1

q_1

k_1

v_1

$q_1 \cdot k_1 = 112$

14

0.88

v_1

z_1

Machines

x_2

q_2

k_2

v_2

$q_2 \cdot k_2 = 96$

12

0.12

v_2

z_2

WORKING KH

- Self-Attention

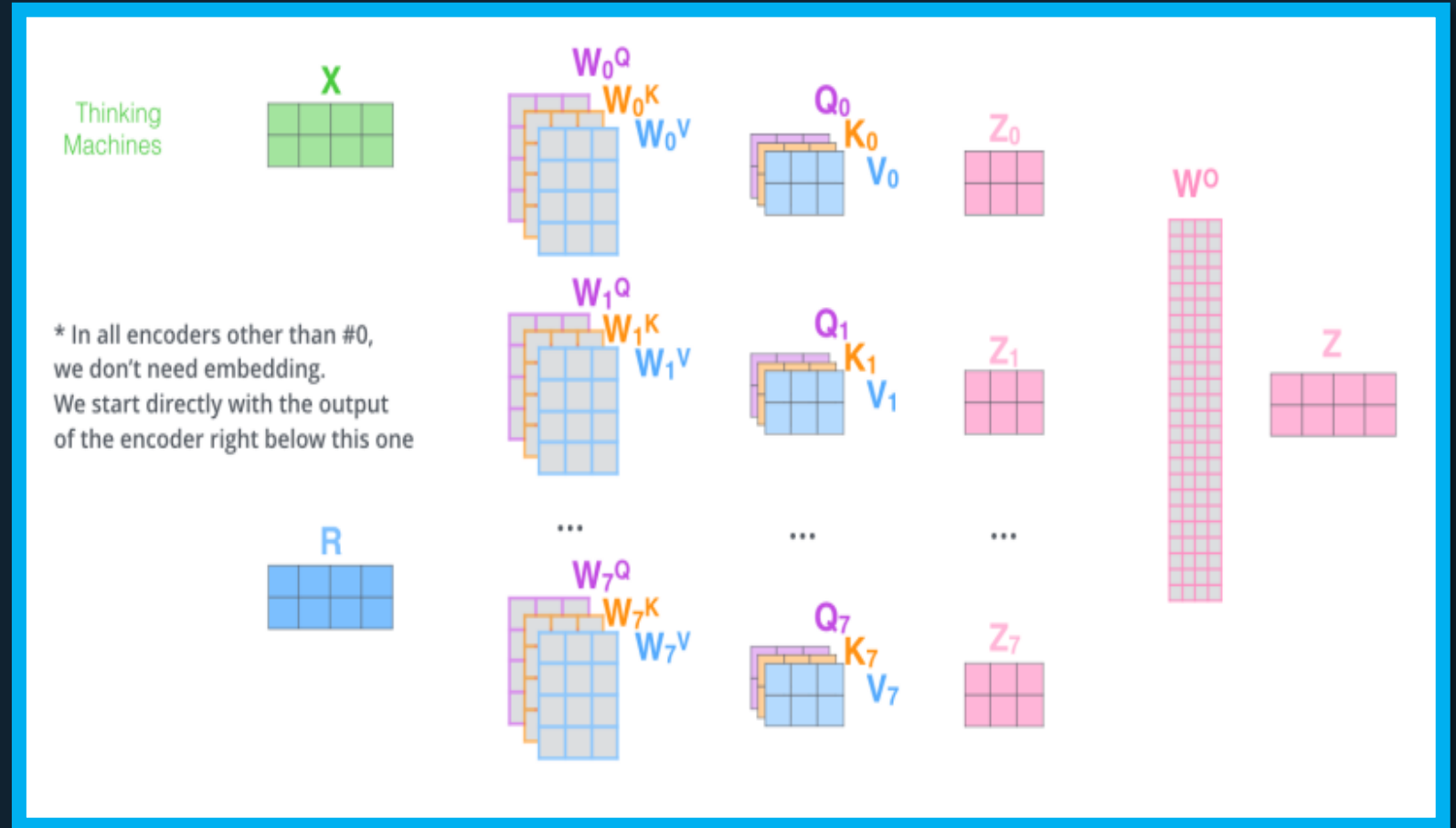
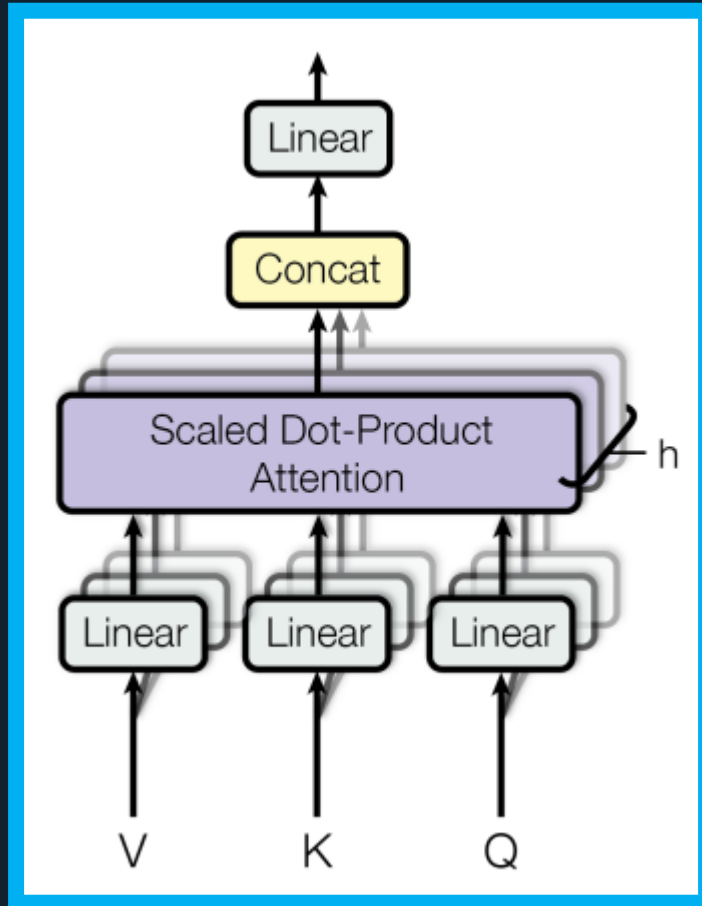
All operations can be calculated as matrix multiplication :

$$\text{softmax}\left(\frac{\begin{matrix} \text{Q} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \times \begin{matrix} \text{K}^T \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix}}{\sqrt{d_k}}\right) \begin{matrix} \text{V} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} = \begin{matrix} \text{Z} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix}$$

$$\begin{aligned} \begin{matrix} \text{X} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \times \begin{matrix} \text{W}^Q \\ \begin{matrix} \text{2x3 grid} \end{matrix} \end{matrix} &= \begin{matrix} \text{Q} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \\ \begin{matrix} \text{X} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \times \begin{matrix} \text{W}^K \\ \begin{matrix} \text{2x3 grid} \end{matrix} \end{matrix} &= \begin{matrix} \text{K} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \\ \begin{matrix} \text{X} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \times \begin{matrix} \text{W}^V \\ \begin{matrix} \text{2x3 grid} \end{matrix} \end{matrix} &= \begin{matrix} \text{V} \\ \begin{matrix} \text{2x2 grid} \end{matrix} \end{matrix} \end{aligned}$$

WORKING KH

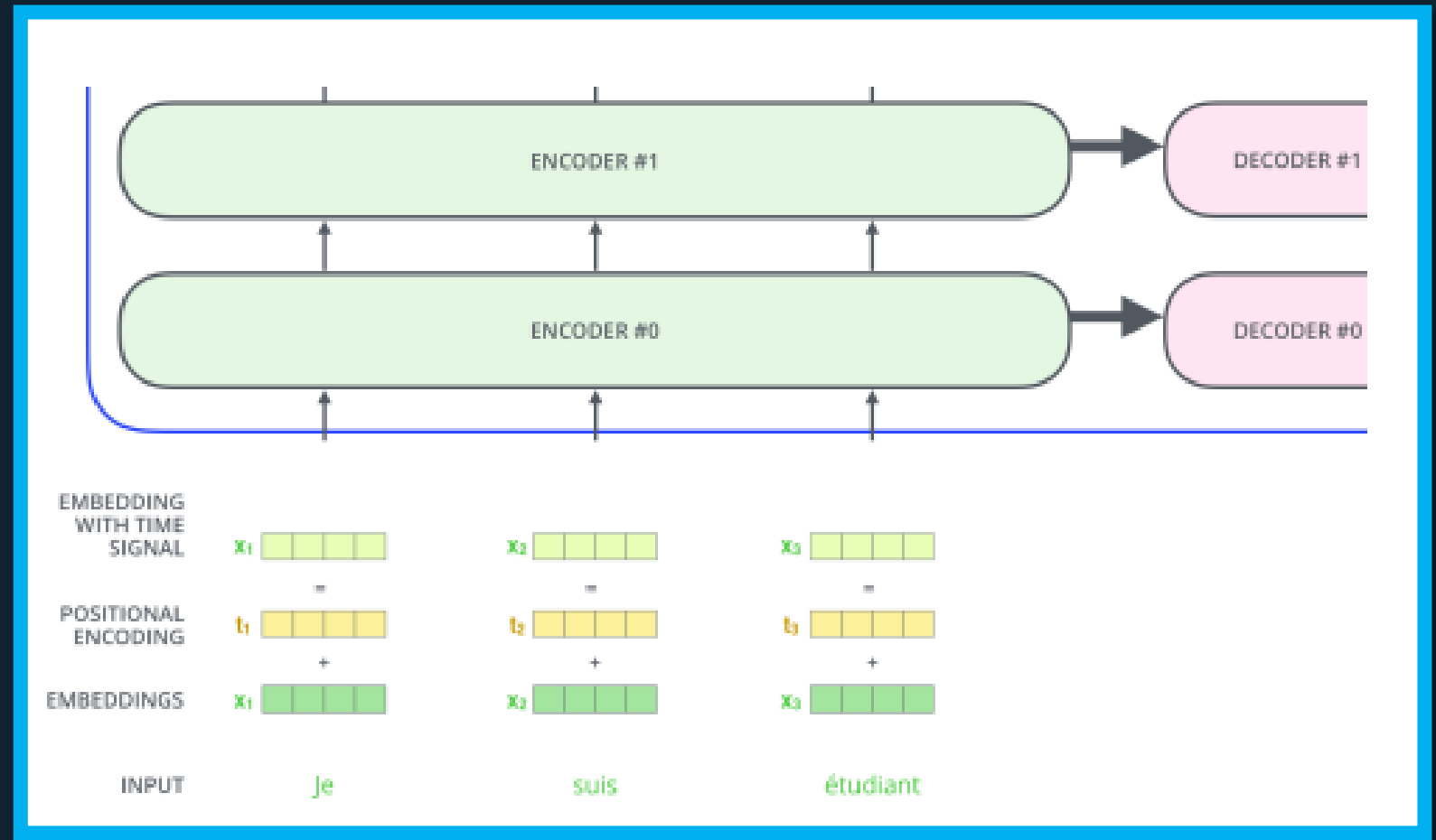
- Multi-Head Attention



WORKING KH

- Positional Encoding

sense of position
follow pattern



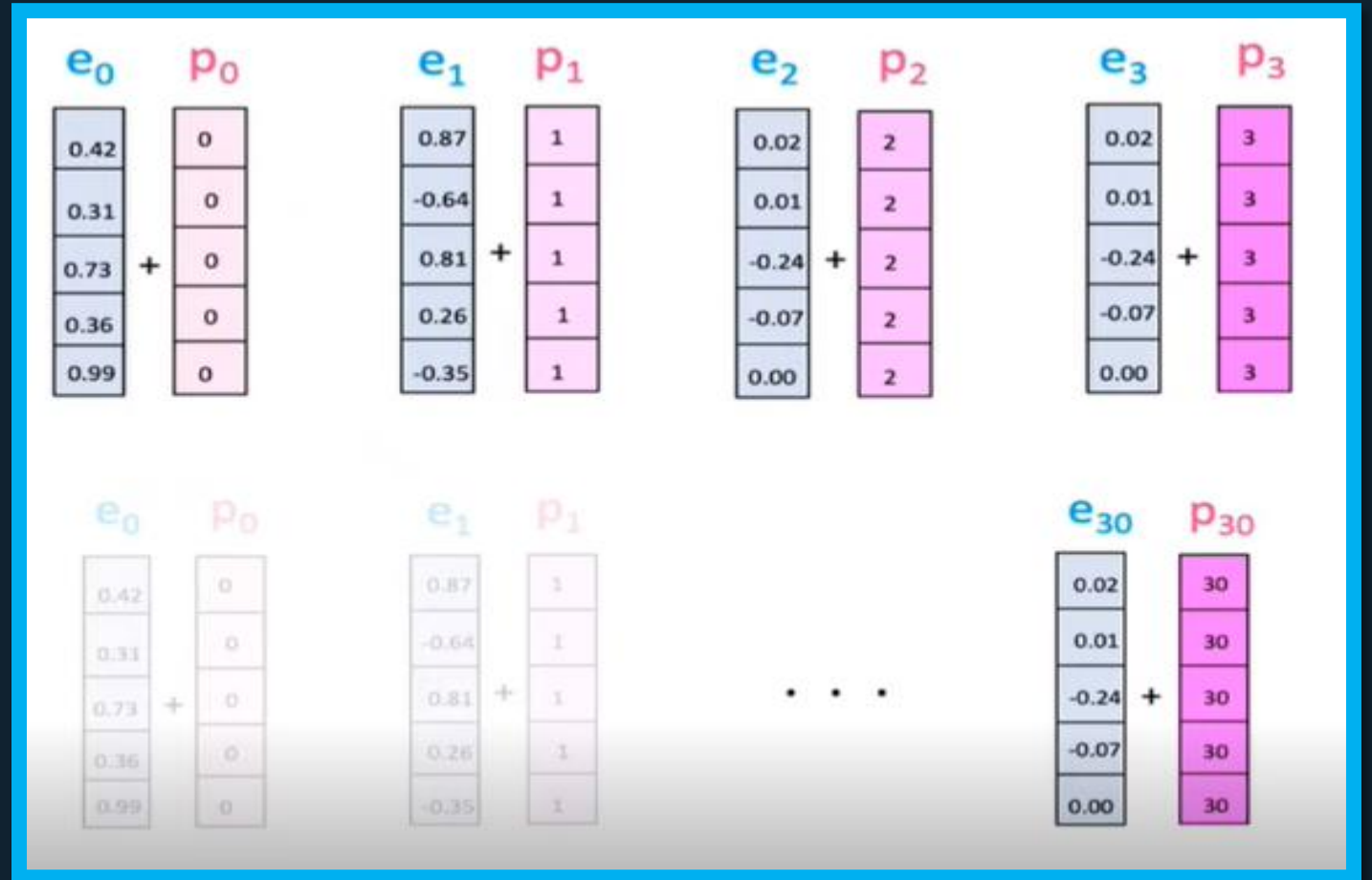
WORKING KH

- Positional Encoding

e_0	p_0	e_1	p_2	e_2	p_1	e_3	p_3
0.42	?	0.87	?	0.02	?	0.02	?
0.31	?	-0.64	?	0.01	?	0.01	?
0.73	?	0.81	?	-0.24	?	-0.24	?
0.36	?	0.26	?	-0.07	?	-0.07	?
0.99	?	-0.35	?	0.00	?	0.00	?

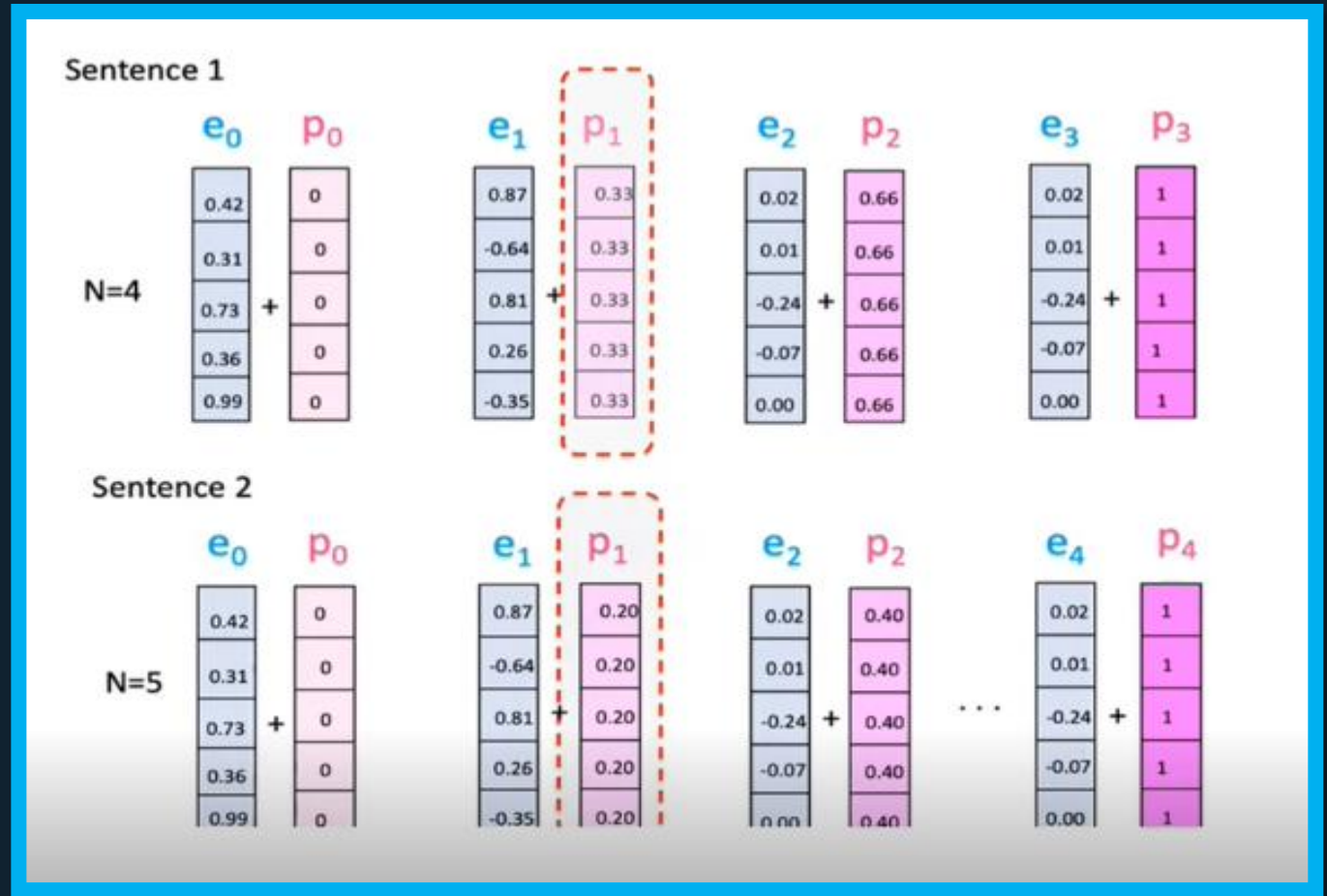
WORKING KH

- Positional Encoding
problem in higher
position tokens



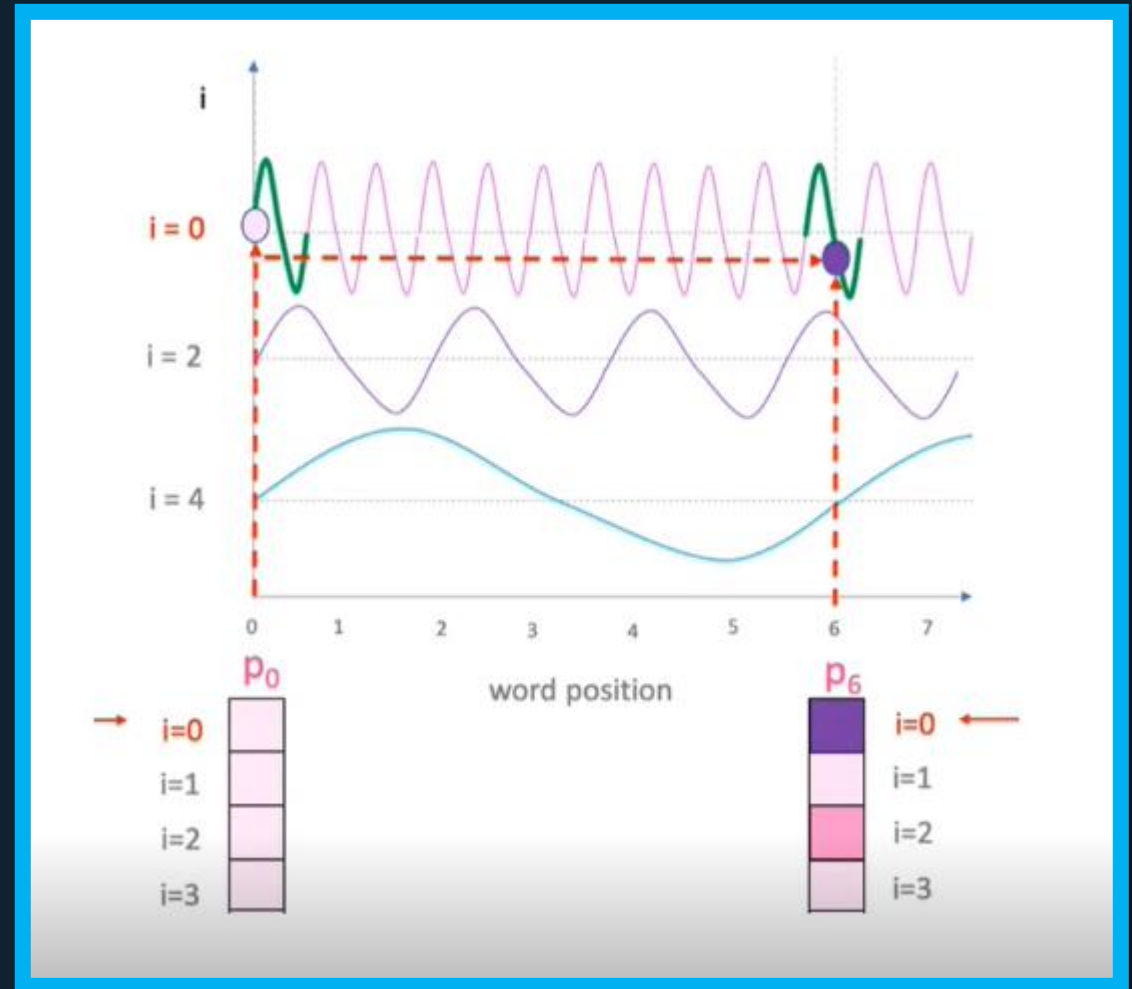
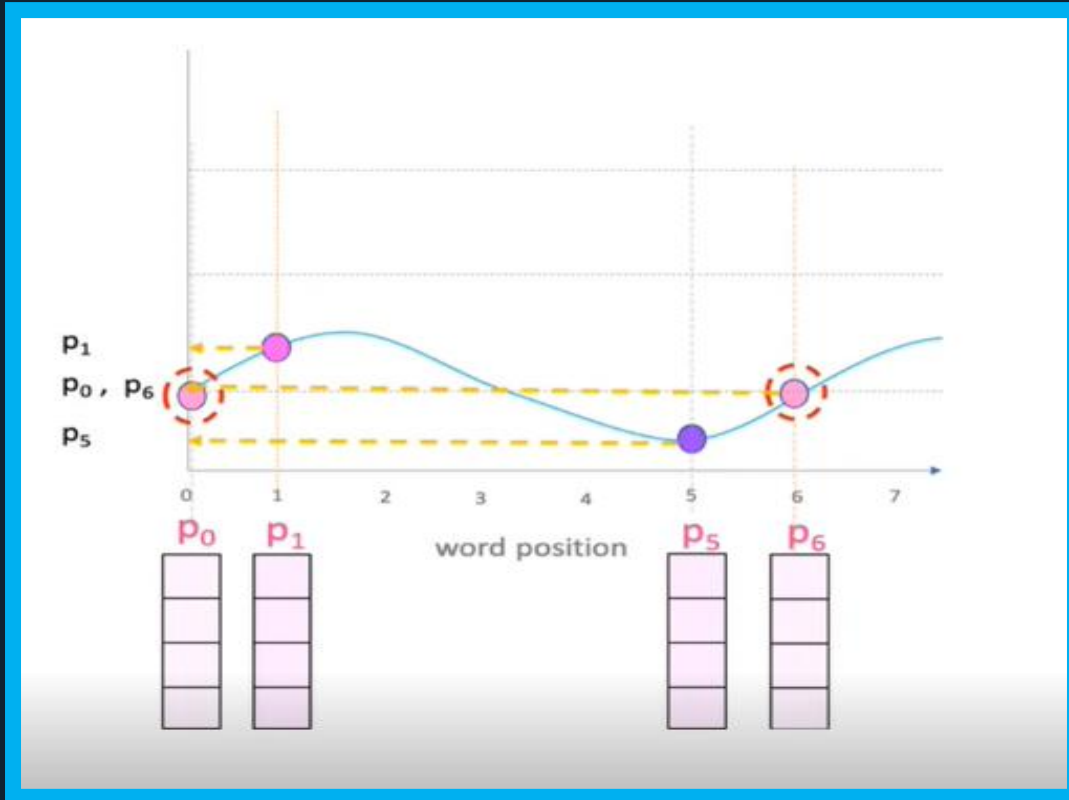
WORKING KH

- Positional Encoding
problem with same
positions in different
sentences lengths



WORKING KH

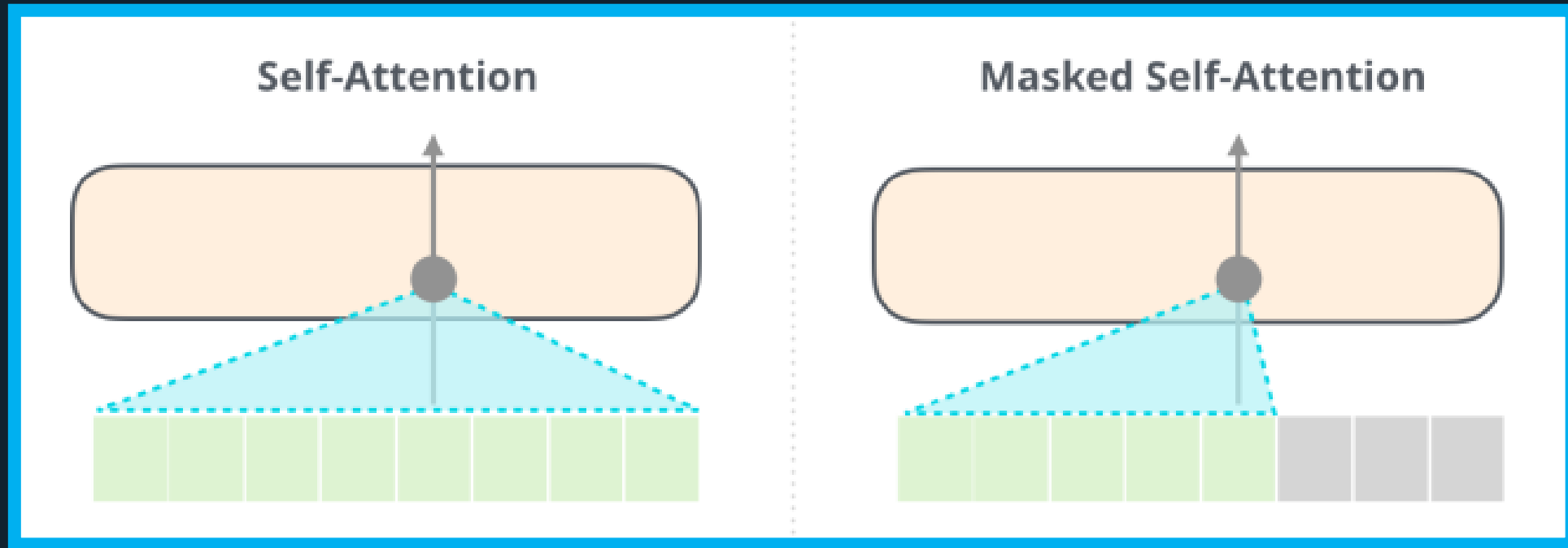
- Positional Encoding



WORKING KH

- Self-Attention

decoder works **exactly** as the encoder **expect** for the mask





THANKS
A LOT